
Learning to negotiate: AI-driven applications versus conventional training

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Abstract: Professional negotiation training plays a substantial role in determining the success of companies in the sales and service field. Sales training courses should offer the up-to-date knowledge and skills needed to respond to changing customer behaviors and desires. Traditional sales training, such as role play, could be cost- and time-consuming. A fully automated, computer-based training application was developed with the help of artificial intelligence (AI). It is designed as a cost-effective substitute and can be adapted for any specific negotiation setting. This research aims to determine whether this application can effectively replace traditional training methods.

Keywords: Negotiation training; sales training; artificial intelligence; natural language processing; technology-supported training application.

1 Introduction

Today, though a number of methods are employed in professional sales training, role-play tends to be the most frequently used (Mani et al. 2016, 184). The role-playing training approach does not provide the same learning experience as selling a real product (Inks et al. 2011, 35). Moreover, this technique can lead to undesired stress, increase nervousness, and, above all, be cost- and time-consuming. The average training effort in

a medium-sized German company corresponds to five full working days per sales employee per year (Binckebanck and Bussmann 2006, 82). While a sales manager needs roughly 50 hours of training per year, entry-level staff need twice as much training as experienced salespeople (Gordon et al. 2012, 666). Although negotiation training has a positive effect on negotiation skills and business success (Movius 2008, 525), authors also report the need for new didactic approaches in teaching those skills (Core et al., 2006, 699; Greco and Murgia 2007, 6).

Technology-supported training applications

Technology-based training applications, including AI and virtual reality (VR), are becoming more accessible and adjustable. This advancing digitalization offers access to perceived-reality learning content (Bues et al. 2018, 113). Several studies focus on the use of AI or VR technologies in the educational field. Lowell and Alshammari (2019, 825) describe a study on experiential learning using AI-driven avatars within online virtual environments. Madani et al. (2010, 15) present a teaching method that uses a software- and AI-based platform. The method enables a small community of players to communicate, interact and play online to learn basic concepts and rules about negotiation and to apply this knowledge. Luo et al. (2021, 14–32) analyze the solutions of AI coaches who use deep-learning algorithms and cognitive speech analysis to assess salespeople's conversations with customers and, in turn, provide training feedback to increase job skills.

Other papers employ VR technology. Broekens et al. (2012, 218–230) report on the test of a VR negotiation training that showed positive effects on the negotiation skills and knowledge. Another paper by Ding et al. (2017, 119–128) reports on a VR negotiation training that exposes users to virtual cognitions while negotiating with virtual characters to improve negotiation knowledge and self-efficacy. Similarly, Taupiac et al. (2019, 1183) present a VR prototype that can be used to train social skills in sales employees.

The relevance of voice assistant technology is also important. Ouali et al. (2017, 1–14) present a voice-controlled agent that can apply different negotiation strategies and negotiate with the user. Pacella et al. (2017, 2–5) present a model of a 3D game based on natural language processing (NLP) that predicts the user's negotiation style. Rosenfeld et al. (2016, 67) present the software application "NegoChat-A," a chat-based negotiation agent introduced as the first application using an NLP module.

Gamification elements are also used to visualize negotiation tools. Greco and Murgia (2007, 2) develop a game-like software application named "Win Win Manager" and demonstrate the added value of this game-based approach.

There is a proven opportunity for technology-based training tools to replicate real learning experiences in professional courses (Liu et al. 2017, 111). Due to the rapid development of those technologies, research on the effectiveness of technology-supported training approaches on performance compared to traditional methods is lacking.

This general problem resonates in the work of Damian et al. (2015, 84-94), where authors present and evaluate a virtual job interview training method that has been adapted to the special requirements of young people with low chances on the job market. The structure of this research is based on the paper of Damian et al. (2015, 84-94).

Research object

The computer-based application introduced in this study was designed by the TriCAT company, a market leader in 3D virtual learning and working environments. It was developed using an AI-supported NLP module. Microsoft Language Understanding Intelligent Services (LUIS) was used for the NLP programming. The version tested in this research was presented on the computer screen, requiring only a microphone and speakers.

The negotiation in the application was modeled according to the phases of the negotiation process: opening, clarifying, bargaining, and closure (Voeth and Herbst 2015, 170). The application is intended to train sales employees or be used in business studies to combine theoretical knowledge with practical exercises. Certain key dates can be pre-set before the application starts. These can be: the purchase object itself, price, payment conditions, delivery period and extra options.

The trainee assumes the role of a professional salesperson through the individual phases. Before the negotiation process starts, the user receives a short briefing from the coach Ringo, an AI-controlled bot that appears on the screen and moderates the whole negotiation process. After the briefing, the trainee negotiates with two virtual, AI-controlled agents, John and Paul, that replicate professional buyers using natural language. The agents proactively lead the dialogue by posing a series of questions. Following each question, the agents wait for the user's response and react accordingly. The system recognizes keywords from the user's spoken language that are then matched to a specific negotiation style. Thanks to the NLP module, the coach Ringo learns the trainee's negotiation techniques. Depending on the wording used, the participant's negotiation style is identified as belonging to one of four categories: accommodating, integrating, avoiding or dominant. After the negotiation process ends, the bot Ringo gives an overview of the used negotiation techniques. He also announces whether the negotiations were successful, depending on the total discount given. The visual representation of the application is on the figure 1.



Figure 1 Professional buyers John and Paul (on the left) and coach Ringo (on the right).

This research aims to determine whether using a fully automated, AI-driven virtual negotiation desktop application could improve negotiation performance in sales situations among undergraduate students with basic knowledge of sales. The study compares the students' negotiation skills, nervousness, and perceived enjoyment using two methods: the computer-based, AI-driven application and traditional role-playing activities. Three research questions are posed:

1. How does overall negotiation performance develop when students use a negotiation role-play approach compared to a technology-supported negotiation application?

2. How are nervousness and perceived enjoyment assessed between training based on a negotiation role play and a technology-supported negotiation application, and how do students assess themselves with respect to these two categories?
3. How does performance develop in terms of content (such as respecting discount limits and non-negotiable negotiation aspects and applying appropriate tactics and negotiation styles) when students use training based on a negotiation role-play approach compared to a technology-supported negotiation approach?

2 Structure of the experiment

This study represents a between-subjects experiment with an experimental and a control group. Regarding the project's overall objective, 27 undergraduate students (13 women and 14 men) from the major of Strategic Sales Management at the Neu-Ulm University of Applied Sciences, presumed to have little or no experience in sales, were recruited for participation in the study. The participants were aged between 21 and 29 years (mean = 23.48; SD = 1.87).

The entire experiment in the form of a training session took place over three days in June 2020. On the first day, all students participated in the initial assessment to provide a baseline for measuring their sales performance. They were randomly assigned to one of four staff members from the assessing university. Three research associates and one professor from the Neu-Ulm University of Applied Sciences, well experienced in negotiation and sales, were trained to conduct the 10-minute interviews. Each assessor evaluated each of their participants with a systematic evaluation form. The students also submitted a self-evaluation form. The forms consisted of questions on performance, nervousness, negotiation techniques, the ability to use discounts, and coping with non-negotiable points (No-Gos). The questionnaire utilized a five-point Likert-type scale (1 [strongly disagree] to 5 [strongly agree]). A higher value indicated a better rating, except for the criterion "nervousness." Here, a lower value was considered a better result. The evaluation questions are listed in Table 1.

Table 1 Evaluation forms for days one and three.

<i>Questionnaire A</i>	<i>Questionnaire B</i>
1) The student's overall performance was optimal.	1) My negotiation performance was optimal.
2) The student was nervous.	2) I was nervous.
3) The student enjoyed him/herself.	3) I enjoyed myself.
4) The student adhered to the discount limits.	4) I adhered to the discount limits.
5) The student did not negotiate on non-negotiables.	5) I did not negotiate on non-negotiables.
6) The student recognized the negotiation tactics of the professional buying agent.	6) I recognized the negotiation tactics of the professional buying agent.
7) The student used different negotiation styles.	7) I used different negotiation styles.

Source: Own representation.

On the second day, the respondents were randomly assigned to the experimental or control group. Each session lasted about 30 minutes. The experimental group received the training in the form of the computer-based application from TriCAT described above. The control group received a role play as a "traditional method." The same evaluators

from day one role played buying agents in simulated negotiations with the students. Both negotiations were modeled according to the phases of the negotiation process: opening, clarifying, bargaining, and closure (Voeth and Herbst 2015, 170). Before starting the experiment, the role-play scenario went through a series of pre-tests to ensure the proper running of the dialogs. The scenario for the role play is presented in Table 2.

Table 2 The role-play scenario.

	I am glad that we will reach a final decision today. Since we last spoke, I have looked at your offer and compared it with the offer made to us by another company. Today our decision will be made: either you will get the contract or your competitor will. So let's not waste time; let's get started. In your offer, we noticed that you allow us an installation date four weeks from today. This surprised us a bit. After all, software installation should always be possible. Your competitor would do the installation for our 300 users as early as next week. Can you offer us something on this point?				
Notes	Dominant ☐	Integrative ☐	Accommodating ☐	Avoiding ☐	Other ☐
	Discount level granted:				
	Recognize negotiation technique:				
	On the payment term, your competitor gives us 60 days. Your offer says 30 days. What can you do for us here?				
Notes	Dominant ☐	Integrative ☐	Accommodating ☐	Avoiding ☐	Other ☐
	Discount level granted:				
	Recognize negotiation technique:	Yes ☐	No ☐		
	No-Go complied:	Yes ☐	No ☐		
	The competitor offers us full service. That means we get 24/7 service. They had only offered us an authority-like service here. What else can you do for us here?				
Notes	Dominant ☐	Integrative ☐	Accommodating ☐	Avoiding ☐	Other ☐
	Discount level granted:	Yes ☐	No ☐		
	Recognize negotiation technique:				
	Now you are still 8% above the competitor's offer. Do you have a final proposal?				
Notes	Dominant ☐	Integrative ☐	Accommodating ☐	Avoiding ☐	Other ☐
	Discount level granted:				
	Recognize negotiation technique:				
	With that, we have clarified all outstanding issues. Thank you very much for coming to us today. We will review the points discussed internally and get back to you by phone today.				

Source: Own representation.

On the third day, the assessment of each participant's negotiation skills from the first day was repeated by the same staff member to compare the performance.

3 Results

Non-parametric tests were conducted (Wilcoxon and Mann–Whitney U tests) because the data lacked a normal distribution. The Mann–Whitney U test was used (unpaired

samples) (Figure 2) for same-day evaluations between different groups. The Wilcoxon test was used (paired samples) (Figure 3) for evaluations for groups on different days.

Research Question 1: How does negotiation performance develop when students use a negotiation role-play approach compared to a technology-supported negotiation application?

On day 1, the Mann–Whitney U test showed no significant differences in terms of “negotiation performance” ($Z = 0.743$; $\text{Sig} = 0.756$) between the assessment results of students who were later assigned to the control group (CG) or the experimental group (EG). The same test on day 3 showed a significant difference in terms of “negotiation performance” ($Z = 0.002$; $\text{Sig} = 0.003$) between the CG and the EG. Researchers used the Wilcoxon test to evaluate results within the groups between day 1 and day 3 and assess the performance improvement of each group. While the CG did not show any differences in terms of “negotiation performance” ($Z = 0$; $\text{Sig} = 1.0$) from day 1 to day 3, our test showed significant differences in the EG with regard to “negotiation performance” ($Z = 1.997$; $\text{Sig} = 0.046$). No significant differences were detected between the groups with respect to the students’ self-assessment on day 1 ($Z = 0.320$; $\text{Sig} = 0.35$) and day 3 ($Z = 0.540$; $\text{Sig} = 0.65$). However, a comparison of different days showed significant within-group differences. “Negotiation performance” ($Z = 2.830$; $\text{Sig} = 0.005$) increased significantly on day 3 within the CG. The EG showed no significant differences ($Z = 1.459$; $\text{Sig} = 0.145$).

Research Question 2: How are nervousness and perceived enjoyment assessed between training based on a negotiation role play and a technology-supported negotiation application, and how do students assess themselves with respect to these two categories?

On day 1, the Mann–Whitney U test showed no significant differences regarding “nervousness” ($Z = 0.456$; $\text{Sig} = 0.488$) and “enjoyment” ($Z = 0.664$; $\text{Sig} = 0.685$) between the assessments of students who were later assigned to the CG or EG. On day 3, a positive trend for the EG was found in the categories “nervousness” ($Z = 0.108$; $\text{sig} = 0.202$) and “enjoyment” ($Z = 0.423$; $\text{sig} = 0.458$). The Wilcoxon test was used to evaluate within-group results between day 1 and day 3 to assess the performance improvement of each group. In the CG, significant differences were determined regarding “nervousness” ($Z = -2.412$; $\text{sig} = 0.016$) and “enjoyment” ($Z = -1.994$; $p = 0.046$), although the effects were smaller than those in the EG. Our tests showed significant differences in the EG (Figure 3 [right]) regarding “nervousness” ($Z = -2.676$; $\text{sig} = 0.007$) and “enjoyment” ($Z = -2.021$; $\text{sig} = 0.043$). Table 3 provides an overview of the median values of the staff assessment and the self-assessment on day 1 and day 3 for the CG and the EG. No significant differences were detected in the students’ self-assessment between the groups on day 1 ($Z = -0.754$; $\text{Sig} = 0.488$) and day 3 ($Z = -0.214$; $\text{Sig} = 0.867$). However, a comparison of different days showed significant within-group differences. The category “nervousness” ($Z = -2.754$; $\text{Sig} = 0.006$) increased significantly on day 3 in the EG. The CG showed no significant differences ($Z = -2.676$; $\text{Sig} = 0.007$) in this category.

Research Question 3: How does performance develop in terms of content (such as respecting discount limits and non-negotiable negotiation aspects and applying appropriate tactics and negotiation styles) when students use training based on a role-play approach, compared to when a technology-supported approach?

On day 1, the Mann–Whitney U test showed no significant differences in terms of discount limits ($Z = 0.504$; $\text{Sig} = 0.650$), non-negotiables ($Z = 0.495$; $\text{Sig} = 0.650$), tactics ($Z = 0.880$; $\text{Sig} = 0.905$), and styles ($Z = 0.393$; $\text{Sig} = 0.458$) between the assessments of students who were later assigned to the CG or EG. On day 3, the same test showed a

positive trend for using tactics ($Z = 0.146$; $\text{Sig} = 0.169$) and styles ($Z = 0.046$; $\text{Sig} = 0.061$). No changes were detected in “discount limits” ($Z = 0.433$; $\text{Sig} = 0.550$) and “non-negotiables” ($Z = 0.165$; $\text{Sig} = 0.550$) because those starting points were already at the highest level (5). The Wilcoxon test was used to evaluate within-group results between day 1 and day 3 to assess the performance improvement of each group. In the CG, “tactics” ($Z = -0.264$; $\text{Sig} = 0.792$) showed an increasing trend from day 1 to day 3, and the trend for “styles” ($Z = -2.598$; $\text{Sig} = 0.009$) was significantly different from day 1 to day 3. No changes were seen in “discount limits” ($Z = -0.378$; $\text{Sig} = 0.705$) and “non-negotiables” ($Z = 0$; $\text{Sig} = 1,0$) because those starting points were already at the highest level (5). In the EG, “tactics” ($Z = -2.020$; $\text{Sig} = 0.043$) and “styles” ($Z = -3.125$; $\text{Sig} = 0.002$) increased significantly. No changes were seen in “discount limits” ($Z = -0.138$; $\text{Sig} = 0.890$) and “non-negotiables” ($Z = 1,732$; $\text{Sig} = 0.083$) because those starting points were already at the highest level (5). Students’ self-assessments showed no significant differences in the between-day comparison or in the comparison of different groups; the “negotiation styles” category was the only exception. “Negotiation styles” ($Z = 2.555$; $\text{Sig} = 0.011$) increased significantly on day 3 in the EG.

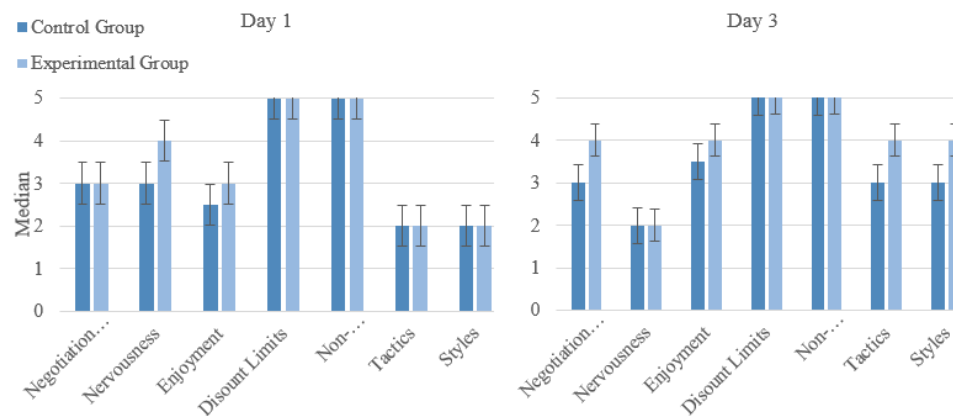


Figure 2 Staff assessment on day 1 (left) and day 3 (right) in the comparison of the control group (CG) and experimental group (EG) based on the median value.

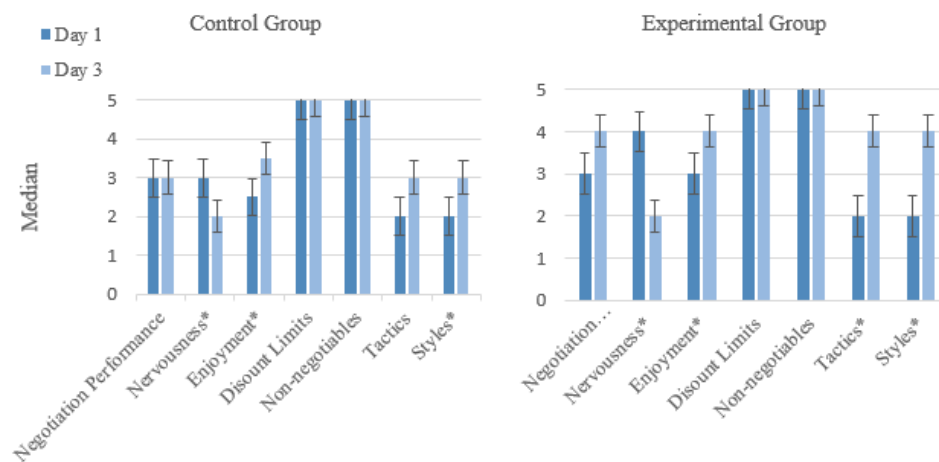


Figure 3 Staff assessment of performance improvement within the CG (left) and EG (right) on day 1 and day 3 based on the median value. Categories marked with * indicate a significant difference.

Table 3 The median values of CG and EG on the first and third days. Significant differences detected between the groups on a specific day are in bold and marked with *. Significant differences within the groups and between the selected days are in italics and marked with ‡.

Staff assessment	Day 1		Day 3		Self-assessment	Day 1		Day 3	
	CG	EG	CG	EG		CG	EG	CG	EG
Negotiation performance	3.00	3‡	3.00	4‡*	Negotiation performance	3‡	3.00	4‡	4.00
Nervousness	3‡	4‡	2‡	2‡	Nervousness	2.00	3‡	1.50	1‡
Enjoyment	2.5‡	3‡	3.5‡	4‡	Enjoyment	4.00	3.00	4.00	4.00
Discount limits	5.00	5.00	5.00	5.00	Discount limits	5.00	5.00	5.00	5.00
Non-negotiables	5.00	5.00	5.00	5.00	Non-negotiables	5.00	5.00	5.00	5.00
Tactics	2.00	2‡	3.00	4‡	Tactics	3.00	3.00	3.00	3.00
Negotiation styles	2‡	2‡	3‡	4‡*	Negotiation styles	2‡	2‡	3‡	4‡

Source: Own representation.

4 Discussion

Analysis of the evaluation forms revealed no significant differences on the first day. The students performed well regardless of whether they were later assigned to the CG or the

EG. Therefore, the progress on day 3 could be attributed to the training on day 2. Both groups (CG and EG) improved between day 1 and day 3. This is unsurprising because all students explored the topic of “negotiation” during the three days. Both methods significantly reduced nervousness and increased enjoyment during negotiation activities. Furthermore, both groups showed significantly increased use of appropriate negotiation styles, though this significance was greater in the EG than in the CG.

Despite both groups’ significant improvements between day 1 and day 3, significant improvements within the EG influenced all categories (including optimal negotiation performance and tactics), excluding discount limits and adherence to non-negotiables. The latter two categories already received the highest score on the Likert scale on day 1 and thus could not be further increased. **This suggests that the technology-supported training system had a greater effect on the students than the conventional role-play method.** Comparing the two groups regarding improvements between day 1 and day 3 (difference analysis) showed that the EG achieved significantly larger improvements in optimal negotiation performance and negotiation styles than the CG.

The student self-assessment revealed some interesting developments. While the students in the CG experienced significant positive developments in terms of negotiation performance and negotiation styles, they did not evaluate their development regarding reduced nervousness as strongly as the staff assessment did. Similarly, they did not record any increase in enjoyment. The students in the EG showed improved negotiation performance and enjoyment, although these improvements were not significant. In contrast, nervousness was reduced significantly. The staff assessment showed significant positive development of negotiation styles. Neither group showed differences in discount limits, non-negotiables, or tactics.

The developments regarding the self-assessment of nervousness and enjoyment showed that the technology-supported training application offered students a more pleasant means of training their negotiation skills than conventional role playing. The students also appeared to enjoy interacting with the application. They were interested in learning how the system analyzed and scored their language use. Some students in the CG wanted to try out the new training application after the experiment. One student provided the following feedback: “To me, it seemed that the negotiation was smoother and much more relaxed. It appears that AI training has proven its positive effect after all! It was fun, too!”

5 Conclusion

This study compared a conventional role-playing method with negotiation training based on AI-based language recognition. **The 3-day user-based study showed that the student group that used the AI-supported training system showed significantly greater improvement regarding negotiation performance, both in terms of the overall picture and the selected sub-criteria, than the students who used conventional role play.** The assessment showed a statistically significant improvement in all possible categories. In addition to the staff assessment conducted by examiners, the students’ self-assessment also showed that they had become more “comfortable” with the technology-supported application than with the conventional role-play method. This paper shows the clear advantages of using this modern technology.

Limitation

This study has some limitations. First, there is a limited sample size. Future research with a broader group of participants is needed. Second, there is a certain degree of subjectivity in respondents' assessment because the same evaluator carried out the initial assessment, role-playing negotiations in the CG and the final evaluation.

Compliance

This study was part of a funded project of the Innovative University program of the Federal Ministry of Education and Research (BMBF) and the federal states of Germany. The study has no conflicts of interest or competing interests. Neither the authors nor the TriCAT company had a special interest in the results of this article. This study was carried out with the intention of testing a technological innovation. It has no medical background and, therefore, no ethical approval was required. Verbal informed consent was obtained prior to the interviews. All participants have consented to the submission of the case report to the journal.

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